MARYAM REZAEE

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A computer science researcher specialising in LLM Interpretability and Cognitively-Inspired AI. My objective is to develop robust, explainable systems by investigating the integration of human inductive biases and symbolic reasoning structures into neural models. My foundational experience includes a thesis on LLM input-output interactions and research in concept-based interpretability. I am now expanding this focus towards neurosymbolic architectures, evidenced by projects in program generation. I seek to advance this research with the aim of developing models capable of more robust and human-aligned reasoning.

Education

M.Sc. in Computer Science — Sep 2023 – Present

Department of Mathematical Sciences, Sharif University of Technology, Iran — AVG: 18.57 / 20.00

Thesis: *Interpretability in Generative Models: Investigating the Mechanisms Behind Output Generation in Large Language Models*

B.Sc. in Computer Science — Sep 2019 – Sep 2023

Department of Mathematical Sciences, University of Isfahan, Iran — AVG: 16.66 / 20.00

Project: *Understanding the Application of Machine Learning in Cell Signalling Pathway Analysis*

Research Experience

Exploring Neurosymbolic AI and Cognitive Architectures — Jul 2025 – Present

Independent Researcher

Sharif University of Technology, Tehran, Iran

- Conducting a self-directed literature review on the integration of neural networks with symbolic reasoning systems to inform future research.
- Studying key topics including program synthesis, logic-based model integration, and the role of human inductive biases in cognitive architectures.
- Implemented a seq2seq model for program generation as a practical exploration of neurosymbolic principles.

Concept-Based Interpretability for Retrieval-Augmented Generation (RAG) Systems on Large Language Models — May 2025 – Present

Research Mentor under the supervision of Dr. Fatemeh Seyyedsalehi

Trustworthy and Generative Machine Learning (TGML) Lab, Sharif University of Technology, Tehran, Iran

- Mentoring a B.Sc. student in developing a novel framework for RAG interpretability.
- Investigating the application of Concept Bottleneck Models (CBMs) to explain the internal generation processes.
- Developing methods using Concept Activation Vectors (CAV) and Automatic Concept-based Explanations (ACE) to enhance the transparency and traceability of RAG outputs.

A Concept Level Energy-Based Framework for Interpreting Black-Box Large Language Model Responses

Link Feb 2025 – Present

Graduate Researcher under the supervision of Dr. Fatemeh Seyyedsalehi

TGML Lab, Sharif University of Technology, Tehran, Iran

- Proposed a model-agnostic, post-hoc framework (ESCI) to interpret black-box LLMs accessed only via APIs.
- Designed an energy model to quantify the conceptual links between user prompts and LLM-generated responses.
- Trained an efficient interpreter network to provide sentence-level importance scores.
- Established proof-of-concept results and currently conducting extensive experimental validation to refine the methodology for submission to ACL; for preliminary preprint, view the above link.

Interpretability in Generative Models: Investigating the Mechanisms Behind Output Generation in Large Language Models

Sep 2024 – Present

Master's Thesis Researcher under the supervision of Dr. Fatemeh Seyyedsalehi

TGML Lab, Sharif University of Technology, Tehran, Iran

- Conducted a comprehensive review of intrinsic, post-hoc, concept-based, and mechanistic interpretability methods for generative models, with a primary focus on LLMs.
- Developed a post-hoc framework to interpret instance-based input-output attribution in LLM generation.
- Analysing internal model mechanisms and their influence on output generation.
- Investigating ways to leverage interpretability findings to improve model robustness, alignment, and performance.

Understanding the Application of Machine Learning in Cell Signalling Pathway Analysis and Developing a DEG Analysis Handbook

Link Feb 2023 – Sep 2023

Bachelor's Project Researcher under the supervision of Dr. Fatemeh Mansoori

University of Isfahan, Isfahan, Iran

- Conducted a literature review on the application of ML and DL (e.g., SVMs) in bioinformatics.
- Analysed and documented methodologies for Gene Expression and Differentially Expressed Gene (DEG) analysis.
- Authored a technical handbook and implementation roadmap for future students to use in DEG analysis.

Foundational Research in Cognitive Neuroscience

Aug 2016 – Feb 2017

Team Researcher (1st Kashan Neuroscience Workshop for Students)

Kashan University of Medical Sciences, Kashan, Iran

- Achieved 2nd Place in a multi-stage research competition for a portfolio on neuroscience and psychology.
- Researched and presented on cognitive topics, including the neural basis of creativity (divergent thinking), hypnotic analgesia, and psychological models of learning/cognitive styles.
- Conducted foundational reviews on core neuroanatomy (Limbic System), the brain-mind problem, and applied clinical neuroscience as part of the workshop curriculum.

Three Centuries in Pursuit of the Theory of Everything

Oct 2015 – Feb 2016

Independent Researcher (Khawrazmi Youth Award)

Farzanegan School (NODET: National Organization for Development of Exceptional Talents), Kashan, Iran

- Authored a comprehensive research paper on the historical and theoretical development of a unified field theory in physics, analysing key milestones from classical mechanics to modern physics.
- Awarded 2nd Place in the Physics category at the Khawrazmi Youth Award (Kashan regional).

Teaching Experience

Relevant to Academic Major

- **Head Teaching Assistant: *Generative Models*** *(M.Sc.)* — Sep 2025 – Present
  Dr. Fatemeh Seyyedsalehi, Sharif University of Technology
- **Teaching Assistant: *Machine Learning Operations*** *(B.Sc.)* — Sep 2025 – Present
  Dr. Fatemeh Seyyedsalehi, Sharif University of Technology
- **Head Teaching Assistant: *Machine Learning*** *(B.Sc.)* — Sep 2025 – Present
  Dr. Mohsen Alambardar, University of Isfahan
- **Instructor: *Fundamentals of Technological Product Development*** — Jul 2024 – Sep 2024
  Tehran University of Medical Sciences
- **Head Teaching Assistant: *Advanced Programming*** *(B.Sc. Math & CS)* — Jan 2023 – Jun 2023
  Dr. Fatemeh Mansoori, University of Isfahan
- **Head Teaching Assistant: *Operating Systems*** *(B.Sc.)* — Jan 2023 – Jun 2023
  Dr. Mojtaba Rafiee, University of Isfahan
- **Instructor: *Algorithms and Data Structures*** *(B.Sc.)* — Nov 2022 – Feb 2023
  Private Classes, University of Isfahan
- **Instructor: *Python Programming*** *(Elementary to Advanced)* — Sep 2019 – Feb 2023
  Private Classes, University of Isfahan

Additional Topics

- **Instructor and Mentor: *Graphic Design with Adobe Illustrator*** — Jan 2023 – Jun 2023
  Academic Association of Mathematics and Computer Science (AMCSUI), University of Isfahan
- **Instructor: *English Language*** *(Intermediate to Advanced)* — Sep 2016 – Nov 2023
  Private Classes

Selected Course Projects

- **System 2 AI** (M.Sc.) — Feb 2025 – Jun 2025
  Dr. M.H. Rohban & Dr. M. Soleymani, Sharif University of Technology — Audit
  PY Program Generation w/ Seq2Seq Models | PY Symbolic Regression | PY GraphRAG | PY LLM Agent
- **Deep Learning** (M.Sc.) — Feb 2025 – Jun 2025
  Dr. Fatemeh Seyyedsalehi, Sharif University of Technology — Mark: 20.00 / 20.00
  PY GPT2 | PY Math Reasoning in LLMs | PY PINN | PY Atari DQN | PY SSL & Contrast. Learning | PY FNN from Scratch

Generative Models (M.Sc.) Sep 2024 – Feb 2025
Dr. Fatemeh Seyyedsalehi, Sharif University of Technology Mark: 18.00 / 20.00
PY RNN vs Transformer · PY Diffusion/DDPM · PY EBM · PY β-VAE · PY Pix2Pix · PY Norm. Flows · PY Bayesian Network

Machine Learning (M.Sc.) Sep 2024 – Feb 2025
Dr. Ali Sharifi Zarchi, Sharif University of Technology Mark: 20.00 / 20.00
PY LoRA from Scratch · PY Word2Vec · PY Knowl.Distil. (KD) & CLIP · PY MobileNet, Eff. & KD. · PY VAE Img Coloring

Matrix Computations (M.Sc.) Sep 2023 – Feb 2024
Dr. Mohammad Reza Razvan, Sharif University of Technology Mark: 19.00 / 20.00
PY SVD RGB-Image Compressor from Scratch · MATLAB Comparing Linear Solvers Implemented from Scratch

Computer Networking (B.Sc.) Feb 2023 – Jun 2023
Dr. Mohsen Alambardar, University of Isfahan Mark: 18.75 / 20.00
PY Group Chat Application with TCP Protocol and Highspeed Data Sharing

Artificial Intelligence (B.Sc.) Sep 2022 – Feb 2023
Dr. Fatemeh Mansoori, University of Isfahan Mark: 19.50 / 20.00
PY AI Search Algorithms for Solving Mazes · PY MDP Gridworld Solver with Value & Policy Iteration

Cryptography (B.Sc.) Sep 2022 – Feb 2023
Dr. Mojtaba Rafiee, University of Isfahan Mark: 19.50 / 20.00
Feasibility of Proving Authenticity and Data Integrity for P2P Using Hash

Fundamentals of Operating Systems (B.Sc.) Feb 2022 – Jun 2022
Dr. Mojtaba Rafiee, University of Isfahan Mark: 19.50 / 20.00
PY Reader-Writer Lock Implementation with Three Priorities

Mathematical Databases (B.Sc.) Feb 2022 – Jun 2022
Dr. Fatemeh Mansoori, University of Isfahan Mark: 19.60 / 20.00
PostgreSQL University Course Registration System · PostgreSQL Academic Transcript Generator & Analyzer

Mathematical Software (B.Sc.) Feb 2022 – Jun 2022
Dr. Najmeh Hoseini, University of Isfahan Mark: 17.75 / 20.00
MATLAB Audio-Signal Analysis · MATLAB CLI University Portal for Grade Management

Algorithms & Data Structures (B.Sc.) Feb 2022 – Jun 2022
Dr. Jaafar Almasizadeh, University of Isfahan Mark: 20.00 / 20.00
PY Chinese Postman · PY Huffman Compressor · PY Max-Flow Altered · PY Prime Sieves Anal. · PY Topol. Sort Anal.

Advanced Programming (B.Sc.) Feb 2020 – Jun 2020
Dr. Jaafar Almasizadeh, University of Isfahan Mark: 20.00 / 20.00
PY RSA Cryptosystem Implementation from Scratch · PY CLI Banking System

Selected Talks

- **ReAGent: A Model-Agnostic Feature Attribution Method for LMs** Link Aug 2025
  Trustworthy & Generative ML (TGML) Lab, Sharif University of Technology
- **Interpreting Language Models with Contrastive Explanations** Link Jun 2025
  TGML Lab, Sharif University of Technology
- **Propagating Universal Perturbations to Attack LLM Guard-Rails** Link May 2025
  Deep Learning Seminars, Sharif University of Technology
- **Universal and Transferable Adversarial Attacks on Aligned LMs** Link May 2025
  Deep Learning Seminars, Sharif University of Technology
- **An Exploration of Concept-Based Interpretability Methods** Link Mar 2025
  TGML Lab, Sharif University of Technology
- **DiT: Scalable Diffusion Models with Transformers** Link Feb 2025
  Generative Models Seminars, Sharif University of Technology
- **Saliency-Guided Training: Interpretability and Robustness in DNNs** Link Jan 2025
  Machine Learning Seminars, Sharif University of Technology
- **Model Sparsity Can Simplify Machine Unlearning** Link Dec 2024
  Machine Learning Seminars, Sharif University of Technology
- **A Foray into Bioinformatics: ML in Cell Signalling Pathway Analysis** Link Jun 2023
  Bachelor's Project Presentation, University of Isfahan
- **Key Takeaways from the 19th Iranian Conference on HPE** May 2018
  Invited Speaker, Post-Conference Seminar, Kashan University of Medical Sciences

Technical Skills

Areas of Expertise	Languages / Core Tools	Modules / Frameworks	Technical Software
ML/DL	Python	PyTorch	Illustrator
GMs/LLMs	R	TensorFlow	InDesign
XAI	MATLAB	Scikit-learn	Photoshop
NeSy AI	PostgreSQL	NumPy	Premiere Pro
Reasoning	LaTeX	Pandas	After Effects
RAG	JavaScript	Transformers	Audition
MLOps	HTML/CSS	Diffusers	Figma
Bioinform	Sockets	Threading	Canvas
CogNeuro	Git/Bash	RISC-V	Docker

Leadership & Project Management

- ***Conceptual Prototype Designer*** Jul 2024 – Aug 2024
  "Siba," the Interprofessional Gamified Platform for Intelligent Education and Evaluation of Medical, Nursing, and Pharmacy Students | *16th Scientific Olympiad for Medical Science Students of Iran*
- ***Founder, Manager, Lead Designer*** May 2023 – Mar 2024
  "PsyCity," the National Competetive and Educational Game for Students of Computer Science
- ***Committee Representative*** Apr 2023 – May 2024
  Assembly for the Academic Associations | *University of Isfahan*
- ***Association President, Head of Content Creation Branch*** Oct 2022 – May 2024
  Association for Mathematics & Computer Science (AMCSUI) | *University of Isfahan*
- ***Content Lead*** Mar 2022 – Apr 2022
  "Varpharin," the Gamified Event for Diabetic Patient Management with an Interprofessional Approach | *AMEE Grant Recipient, Presented at AMEE 2023 (Short Communication)*

Professional Activities & Service

- ***Invited AI Panelist*** Aug 2025
  Symposia on AI in Health Professions Education | *International AMEE 2025 Conference, Live*
- ***English Interpreter & Student Task Force*** May 2023
  International Guests & Presenters | *24th Iranian Conference on Health Professions Education*
- ***UI/UX & Graphic Designer*** Since 2018
  Academic & Institute Projects | *Freelance*
- ***Academic & Technical Translator*** Since 2016
  Book & Article Publication | *Freelance*

Awards

Honouree: **Tech & Innovation** 2024
17th International Harekat Festival, Iran

Top: **Best & Most Creative Pres.** 2024
17th University-Level Harekat Festival, UI, Iran

Honouree: **Educational Assoc.** 2023
16th International Harekat Festival, Iran

2nd: **Best Research in Physics** 2016
18th Khawrazmi Youth Award, Kashan, Iran

Top: **Digital Content** 2024
17th International Harekat Festival, Iran

1st: **Most Active Association** 2024
Tadaee Festival, University of Isfahan, Iran

Honouree: **Best Team** 2018
Avicenna Student Creativity Cup, Iran

1st: **Poetry** 2016, 2017, 2018
Regional Cultural-Artistic Cup, Kashan, Iran

Top: **Best Science Assoc.** 2024
17th University-Level Harekat Festival, UI, Iran

Honouree: **Tech & Innovation** 2024
Roshana Festival, Science Ministry, Iran

2nd: **Best Team** 2016
1st Kashan Neuroscience Workshop, Iran

1st: **Short Story** 2016, 2017
Regional Cultural-Artistic Cup, Kashan, Iran

Languages

- Full Proficiency in ***English***
- Native Fluency in ***Persian (Farsi)***